After trading costs, how much money remains?

Keep the one-day return rule from Week 4. Separate cost rates, execution dates and the account after costs; do not select weighting methods.

Issue	Question
01	Can daily prices reveal the cost of our trade?
02	How many dollars does 4 basis points cost?
03	How do three commission schedules charge for the same stock?
04	Why allow half the spread for each direction?
05	When is the spread counted twice?
06	Record the purchase cost on the 9th or the 10th?
07	Does a purchase and a sale each incur a cost?
08	First round trip: how much of USD 1,000 remains?
09	Why subtract 4 basis points to obtain the net return?
10	Why does the terminal wealth gap differ from total fees?
11	How can USD 1,000 cover both stock and purchase cost?
12	How does buying one close later change the first holding return?
13	Can the first worse return determine the entire delayed account?
14	Do fewer trades necessarily leave more money?
15	What happens to terminal wealth when only the cost rate rises?
16	Which cost rate eliminates this historical profit?
17	How do cash, buy-and-hold and the trading rule compare?
18	With a USD 1 minimum per order, does a small order still cost only 4bp?

Which costs are added to the same trading rule?

AAPL, 2024/1/2 to 6/28. Request stock when the Close exceeds the previous trading-day Close; otherwise request cash. Trade at the next trading-day Close.

Start with USD 1,000 cash and allow fractional shares. Exclude dividends, taxes and cash interest. At the final Close, value the account without a new trade or forced sale. Next-close execution is a teaching assumption, not a restriction on U.S. trading.

Predict: does the stock request on 1/8 execute on 1/8 or at the 1/9 Close?

```
import numpy as np
import pandas as pd
import yfinance as yf
aapl = yf.Ticker("AAPL").history(
    start="2024-01-02", end="2024-07-01", interval="1d",
    auto_adjust=False, actions=True)
aapl.index = aapl.index.strftime("%Y-%m-%d")
close = aapl["Close"]
assert close.index.is_unique and close.index.is_monotonic_increasing
assert np.isfinite(close).all() and (close > 0).all()
price_change = close / close.shift(1) - 1
signal = price_change.gt(0).astype(float).where(price_change.notna())
dates = ["2024-01-08", "2024-01-09", "2024-01-10"]
pd.DataFrame({"Observation date": dates,
    "Close (USD/share)": [f"{close[d]:.2f}" for d in dates],
    "After-close request": ["Hold stock" if signal[d] == 1 else "Cash" for d in dates]})
```

Observation date	Close (USD/share)	After-close request
2024-01-08	185.56	Hold stock
2024-01-09	185.14	Cash
2024-01-10	186.19	Hold stock

signal is the after-close request: 1 for stock, 0 for cash. The 1/8 stock request buys at the 1/9 Close; the 1/9 cash request sells at the 1/10 Close. The 124 prices form 123 intervals. The first date has no previous Close; the account starts in cash.

The 4 bp rate per direction is assumed, not estimated from daily prices. Approximate dollar cost by rate times the change in stock weight times opening account value. See Issue 2 for basis points and Issue 11 for the difference from an exact shares-and-cash budget.

ISSUE 01

Can daily prices reveal the cost of our trade?

Separate market prices from order records. A price change alone does not show how much extra an order cost.

Predict: will this table include our order size, execution time and commission?

```
dates = ["2024-01-08", "2024-01-09"]
pd.DataFrame({"Observation date": dates,
    "Open (USD/share)": aapl.loc[dates, "Open"].round(2).values,
    "Close (USD/share)": close.loc[dates].round(2).values})
```

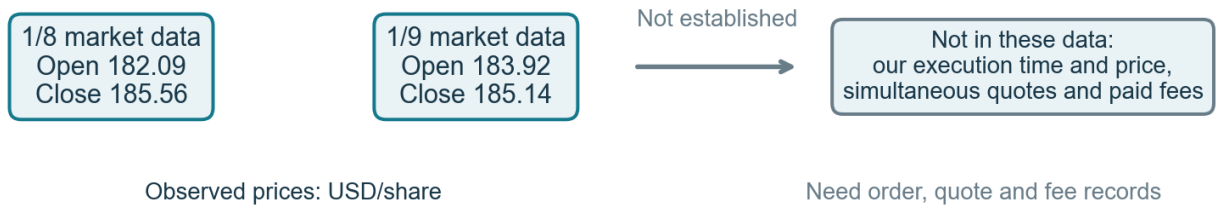
Observation date	Open (USD/share)	Close (USD/share)
2024-01-08	182.09	185.56
2024-01-09	183.92	185.14

The 1/8 Close of 185.56 forms the buy signal; the main example buys at the 9th Close of 185.14. The 9th Open of 183.92 and Close of 185.14 are observations at different times. Their difference is not automatically this order's slippage.

Commission requires a fee schedule; spread requires simultaneous bid and ask quotes. Slippage compares our execution price with a specified reference, such as the midpoint at order submission. This needs our execution record, which daily data do not contain.

Use an assumed 4 bp per direction to study the account effect. These daily prices do not establish actual spreads, slippage or execution prices.

Daily prices are not our execution or cost records



Open-to-Close movement is not this order's slippage. These data provide neither our execution nor simultaneous quotes.

Discuss

To measure slippage for the 1/9 purchase, which execution record is missing from the daily price table?

Sources: [1], [2], [3], [4].

ISSUE 02

How many dollars does 4 basis points cost?

Convert the rate to a decimal, then multiply by the amount traded. A 4 bp rate does not mean charging the whole account every day.

1 basis point (bp) is 0.01 percentage points, or 0.01%. Thus 4 bp = 0.04% = 0.0004. The 1/9 Close is about USD 185.14/share. Assume buying one share at that price solely to calculate the fee.

q is shares, C is price in USD/share, F is cost in USD, and c is the decimal rate. For one share, q=1 and the traded amount is USD 1×185.14.

$$F = cqC, \qquad 0.0004(1)(185.14) \approx 0.074056 \text{ USD}$$

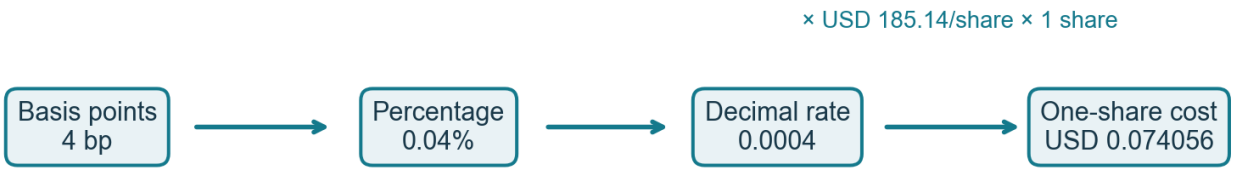
Predict: is the 4 bp cost for one share above or below one cent?

```
price = close["2024-01-09"]
bp = np.array([1, 4, 10, 100])
pd.DataFrame({"Assumed rate (bp)": bp, "Rate (%)": bp/100,
    "One-share fee (USD)": (price*bp/10000).round(5)})
```

Assumed rate (bp)	Rate (%)	One-share fee (USD)
1	0.01	0.01851
4	0.04	0.07406
10	0.1	0.18514
100	1.0	1.8514

4 bp costs about USD 0.07406, or 7.41 cents, not 4%. If only part of the capital trades, charge only that traded amount.

One share on 2024/1/9: convert 4 bp to a decimal first



4 bp is not 4%. At the 1/9 Close, the assumed one-share cost is about USD 0.07406.

Discuss

At the same price, what is the 4 bp dollar cost for ten shares?

Sources: [2], [5].

ISSUE 03

How do three commission schedules charge for the same stock?

A fixed fee depends on order count, a per-share fee on shares, and a proportional fee on traded value. Identify the schedule before comparing costs.

Use the 1/9 price of USD 185.14/share. Assume one order for 1, 10 or 100 shares. Compare USD 1 per order, USD 0.005 per share and 4 bp of traded value. These are alternatives, not additive charges.

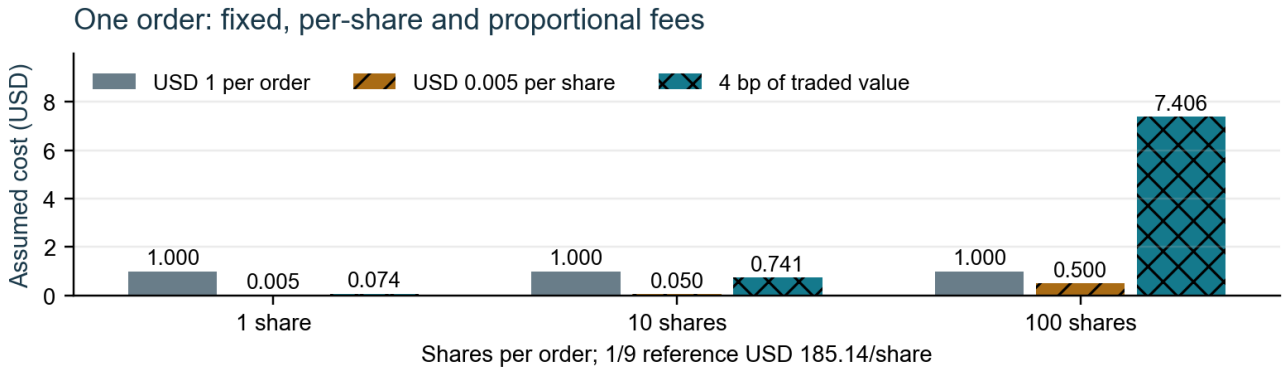
q is shares, C is price per share and F is cost in USD. Ten shares trade about USD 1,851.40; the three costs are USD 1.00, 0.05 and 0.74056. These are hypothetical fee schedules, not a borrowed ten-share order in the main account.

$$F_{\text{fixed}} = 1, \quad F_{\text{share}} = 0.005q, \quad F_{\text{value}} = 0.0004qC$$

Predict: which fee remains USD 1 when shares increase?

```
counts = np.array([1, 10, 100])
pd.DataFrame({"Order shares": counts, "Per-order fee (USD)": 1.0,
    "Per-share fee (USD)": counts*0.005,
    "Proportional fee (USD)": (counts*price*0.0004).round(5)})
```

Order shares	Per-order fee (USD)	Per-share fee (USD)	Proportional fee (USD)
1	1.0	0.005	0.07406
10	1.0	0.05	0.74056
100	1.0	0.5	7.4056



The fixed charge remains USD 1 per order. Per-share cost rises with shares; proportional cost rises with traded value. Do not add the three alternatives.

Discuss

If ten shares are split into ten one-share orders, how much does the USD 1 per-order schedule charge?

Sources: [2], [5].

ISSUE 04

Why allow half the spread for each direction?

The full bid-ask spread spans bid to ask. Relative to the midpoint, each direction covers half.

An immediate purchase normally faces ask; a sale faces bid. Relative to the simultaneous midpoint, buying pays half-spread more and selling receives half-spread less. With this fixed 4 bp full spread at both trades, each direction is 2 bp, totaling 4 bp. Actual quotes can change.

Assume a full spread of 4 bp and allow 2 bp per direction. Use the 1/9 Close of USD 185.14/share only to convert the allowance into dollars, not as an observed quote midpoint.

$$4/2 = 2 \text{ bp}, \qquad 185.14(2/10000) \approx 0.03703 \text{ USD/share}$$

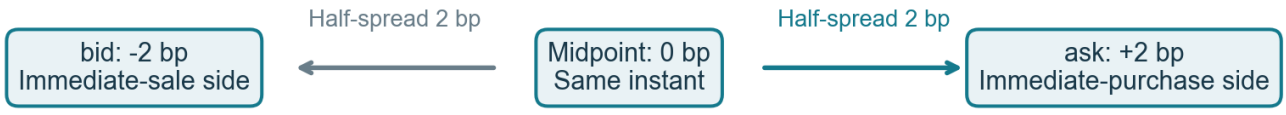
Predict: does the same 2 bp imply the same dollar allowance per share on both dates?

```
dates = ["2024-01-09", "2024-01-10"]
pd.DataFrame({"Reference date": dates,
    "Close (USD/share)": close.loc[dates].round(2).values,
    "2 bp allowance (USD/share)": (close.loc[dates]*.0002).round(5).values})
```

Reference date	Close (USD/share)	2 bp allowance (USD/share)
2024-01-09	185.14	0.03703
2024-01-10	186.19	0.03724

The purchase-date allowance is about USD 0.03703/share; the sale-date allowance is about 0.03724. The rate is unchanged, but dollars vary with the reference price.

Assumed full spread 4 bp: each direction is 2 bp



Dollar conversion only: 1/9 Close × 2 bp = 0.03703; 1/10 = 0.03724 USD/share

The diagram shows assumed relative distances, not actual quotes. Both Close values only convert the allowance into dollars; neither is treated as a midpoint.

Discuss

For a 4 bp full spread, how much extra is counted if both buying and selling are charged 4 bp?

Sources: [3], [4].

ISSUE 05

When is the spread counted twice?

Specify the reference used for slippage before deciding whether it already includes the spread.

Assume commission 1 bp + half-spread 2 bp + additional slippage 1 bp = 4 bp per direction. Additional means beyond the spread. These numbers are not estimated from AAPL daily data.

A total midpoint-to-execution difference of 3 bp already includes half-spread 2 bp and additional slippage 1 bp. Adding commission gives 4 bp; do not add half-spread again.

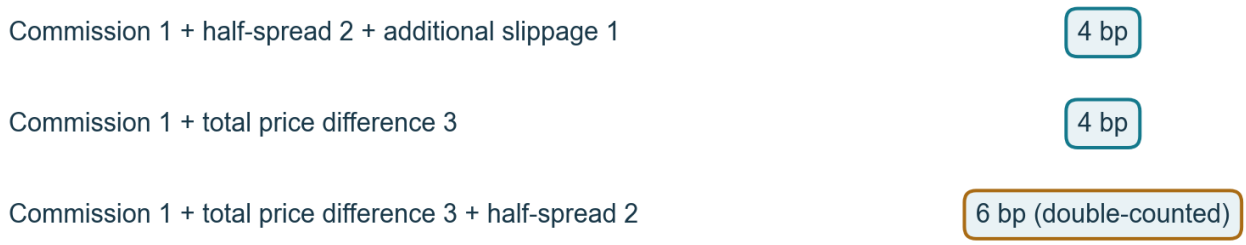
Predict: which row charges the same spread again?

```
rates = np.array([4, 4, 6])
pd.DataFrame({"Method": ["Separate components", "Total price difference", "Spread counted twice"],
    "Rate (bp)": rates,
    "One-share cost (USD)": (price*rates/10000).round(5)})
```

Method	Rate (bp)	One-share cost (USD)
Separate components	4	0.07406
Total price difference	4	0.07406
Spread counted twice	6	0.11108

The first two rows each cost about USD 0.07406; the third costs 0.11108. The extra 2 bp double-counts a component, not an observed worse execution. Actual slippage can be favorable; examine direction, reference and execution price.

Assumed costs: do not add a spread already in the price difference



The total 3 bp difference includes the 2 bp spread component. Adding another 2 bp counts that component twice.

Discuss

If 1 bp slippage is measured from ask rather than midpoint, should the 2 bp half-spread still be added separately?

Sources: [3], [4], [5].

ISSUE 06

Record the purchase cost on the 9th or the 10th?

Buying at the 1/9 Close incurs cost that day. The first holding return ends on the 10th. Do not date both events identically.

Decide to hold stock on the 8th and buy at the 9th Close, 185.14. Hold from the 9th to the 10th, then sell at the 10th Close. held is the stock weight over the ending interval; after is the post-trade weight. Here 1 means all stock, 0 cash, not shares.

Predict: which date incurs purchase cost before earning a holding return?

```
held = signal.shift(2).fillna(0)
after = signal.shift(1).fillna(0)
after.iloc[-1] = held.iloc[-1]
change = after-held
turnover = change.abs()
actions = {1: "Buy", 0: "No trade", -1: "Sell"}
dates = ["2024-01-08", "2024-01-09", "2024-01-10"]
pd.DataFrame({"Observation date": dates,
    "Held weight": held.loc[dates].astype(int).values,
    "Post-trade weight": after.loc[dates].astype(int).values,
    "Action": [actions[int(change[d])] for d in dates]})
```

Observation date	Held weight	Post-trade weight	Action
2024-01-08	0	0	No trade
2024-01-09	0	1	Buy
2024-01-10	1	0	Sell

On the 9th, 0 to 1 means buy; on the 10th, 1 to 0 means sell. after uses the previous signal: decide at the previous Close, execute at this Close. held is one row earlier. Set final after equal to held so 6/28 only values the account. The execution assumption determines these shifts.

One-way turnover here is the absolute change in stock weight across execution: 0→1 or 1→0 gives 1; no trade gives 0. It determines whether to charge 4 bp that day, not a number of shares.

Pay for buying on 1/9; the first holding return ends on 1/10



Record the purchase cost on 1/9 even before earning a stock return. On 1/10, value the holding return and record the separate sale cost.

Discuss

Why must the 1/9 purchase cost not be postponed to the 10th?

Sources: [5], [6].

ISSUE 07

Does a purchase and a sale each incur a cost?

Continuing to hold is not buying again every day. A completed round trip does include two charged directions: purchase and sale.

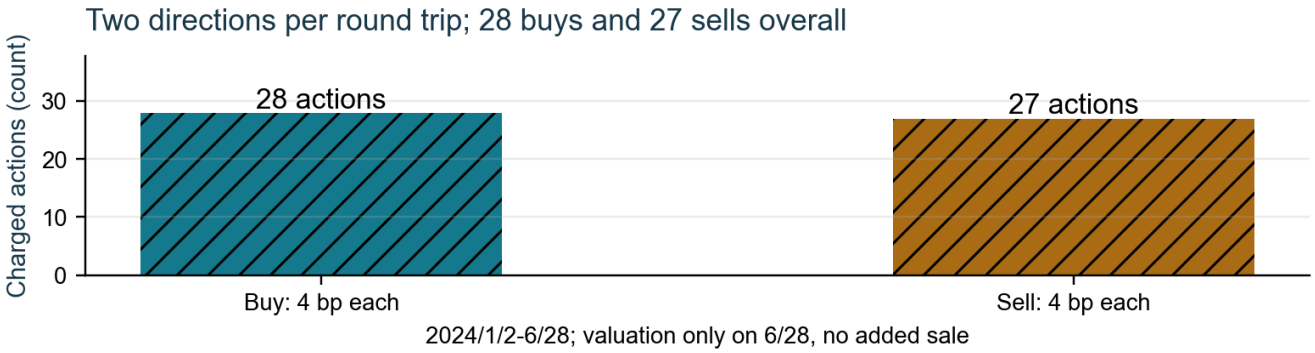
Predict: for buying on 1/9 and selling on 1/10, is 4 bp charged once or twice?

```
trades = pd.DataFrame({"Action": change, "Turnover": turnover})
trades = trades.loc[turnover.gt(0)].copy()
trades["Direction"] = np.where(trades["Action"].gt(0), "Buy", "Sell")
shown = trades.iloc[:2]
pd.DataFrame({"Trade date": shown.index, "Direction": shown["Direction"].values,
    "Cost (bp)": 4*shown["Turnover"].values})
```

Trade date	Direction	Cost (bp)
2024-01-09	Buy	4.0
2024-01-10	Sell	4.0

Charge 4 bp at each trade, but the bases differ. Calculate dollar costs separately (Issue 8), not 8 bp times one common amount.

There are 28 buys, 27 sells and 55 charged actions. On 6/28 stock is valued but not sold, so the final position is not a completed round trip. Do not add a sale merely to equalize buy and sell counts.



The first 1/9 buy and 1/10 sale each cost 4 bp. The period has 55 charged actions; do not add a final sale to complete the last round trip.

Discuss

If all stock is instead sold for cash on 6/28, which additional cost is incurred?

Sources: [2], [5].

ISSUE 08

First round trip: how much of USD 1,000 remains?

Follow capital, holding P/L and both costs from the 1/9 buy to the 1/10 sale.

Before buying on 1/9, there is USD 1,000. Approximate cost is 0.40, leaving USD 999.60. The price rises from 185.14 to 186.19 at the 10th Close: about +0.567140% per share.

$$999.60\left(\frac{186.19}{185.14}-1\right)\approx 5.669 \text{ USD}$$

The main model approximates the sale base by opening wealth 999.60: cost is USD 999.60×0.0004=0.39984, not a charge on the stock value just before selling.

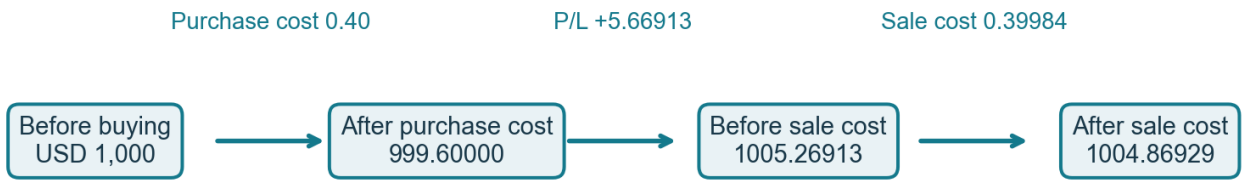
Predict: with no holding P/L on the purchase day, can ending wealth still be below 1,000? All amounts are USD.

```
first_buy_fee = 1000 * 0.0004
first_open = 1000 - first_buy_fee
first_gain = first_open * (close["2024-01-10"] / close["2024-01-09"] - 1)
first_exit_fee = first_open * 0.0004
first_end = first_open + first_gain - first_exit_fee
pd.DataFrame({"Execution date": ["2024-01-09", "2024-01-10"],
    "Opening value": [f"{1000:.5f}", f"{first_open:.5f}"],
    "Holding P/L": ["0.00000", f"{first_gain:.5f}"],
    "Approximate cost": [f"{first_buy_fee:.5f}", f"{first_exit_fee:.5f}"],
    "Ending value": [f"{first_open:.5f}", f"{first_end:.5f}"]})
```

Execution date	Opening value	Holding P/L	Approximate cost	Ending value
2024-01-09	1000.00000	0.00000	0.40000	999.60000
2024-01-10	999.60000	5.66913	0.39984	1004.86929

The table uses full price precision. Substitution with the two-decimal displayed Close values gives an approximation.

Buy 1/9, sell 1/10: linear cost approximation using opening wealth



Amounts are USD; approximate sale cost uses opening 999.60 × 4 bp, not pre-sale market value

Pay USD 0.40 to buy; next day gain 5.66913, then pay 0.39984 to sell, leaving USD 1004.86929.

Pre-sale stock value is about 1005.26913; the main approximate cost 0.39984 leaves 1004.86929. These costs differ from the fixed USD 1 per trade in Week 4.

Discuss

If 4 bp is charged on pre-sale stock value USD 1005.26913 instead, is the cost equal to 0.39984?

Source: [5].

ISSUE 09

Why subtract 4 basis points to obtain the net return?

Divide dollar cost by the same opening wealth before subtracting it from return. Dollars cannot be subtracted directly from a percentage.

w is opening stock weight; r is decimal stock return; c=0.0004. τ is one-way turnover, F cost (USD), and W opening wealth (USD). w and τ are 0 or 1. gross/net mean account returns before/after cost, in decimals.

$$r_{\text{net}} = wr - c\tau, \qquad F/W = c\tau$$

1/10: cost 0.39984 / opening wealth 999.60 = 0.0004 = 0.04%. Holding return 0.567140% - 0.04% = net return 0.527140%.

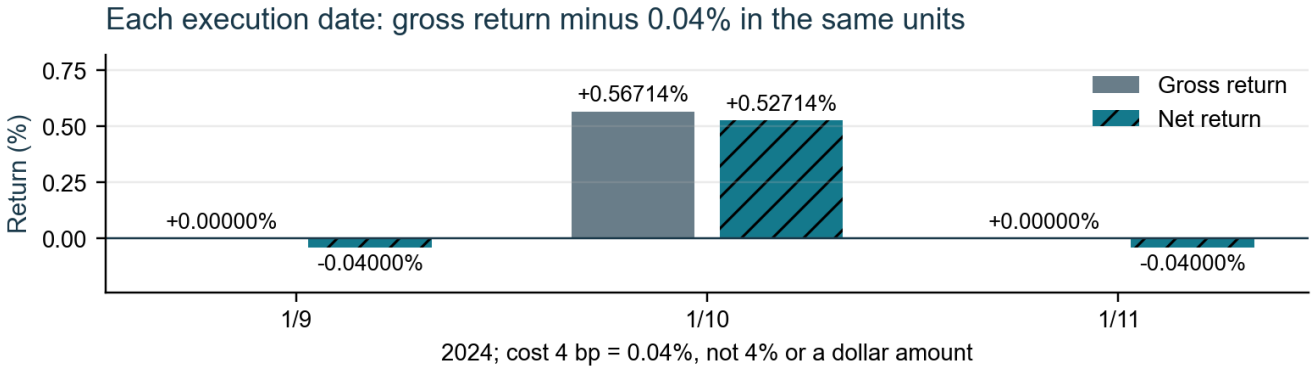
$$\frac{5.66913 - 0.39984}{999.60} \approx 0.00527140 = 0.527140\%$$

Predict: why is the 9th net return negative before earning a stock return?

```
asset_return = close.pct_change(fill_method=None)
gross = asset_return.where(held.eq(1), 0.0)
cost_rate = 0.0004
net = gross-cost_rate*turnover
dates = ["2024-01-09", "2024-01-10", "2024-01-11"]
pd.DataFrame({"Observation date": dates,
    "Holding return (%)": (100*gross.loc[dates]).round(5).values,
    "Cost (bp)": (4*turnover.loc[dates]).values,
    "Net return (%)": (100*net.loc[dates]).round(5).values})
```

Observation date	Holding return (%)	Cost (bp)	Net return (%)
2024-01-09	0.0	4.0	-0.04
2024-01-10	0.56714	4.0	0.52714
2024-01-11	0.0	4.0	-0.04

The 9th and 11th have no holding return, but each purchase costs 0.04%. The 10th has a holding return and also a sale cost.



On 1/9 the account has not held stock: gross return is 0; purchase cost makes net return -0.04%. Cost is also charged once on 1/10 and 1/11.

Discuss

What unit error occurs if USD 0.39984 is subtracted directly from 0.567140%?

Sources: [5], [6].

ISSUE 10

Why does the terminal wealth gap differ from total fees?

Fix the one-day signal, execution dates and period. Compare wealth after sequential 4 bp charges with the model's sum of dollar costs.

Gross means before cost; net means after cost. gross_wealth and net_wealth are account values in USD. fees uses previous-close net wealth ×0.0004×turnover at each trade, not a broker bill. W is dollar wealth; $1+r_{net}$ is the ending/opening factor. The next interval starts from the balance after costs.

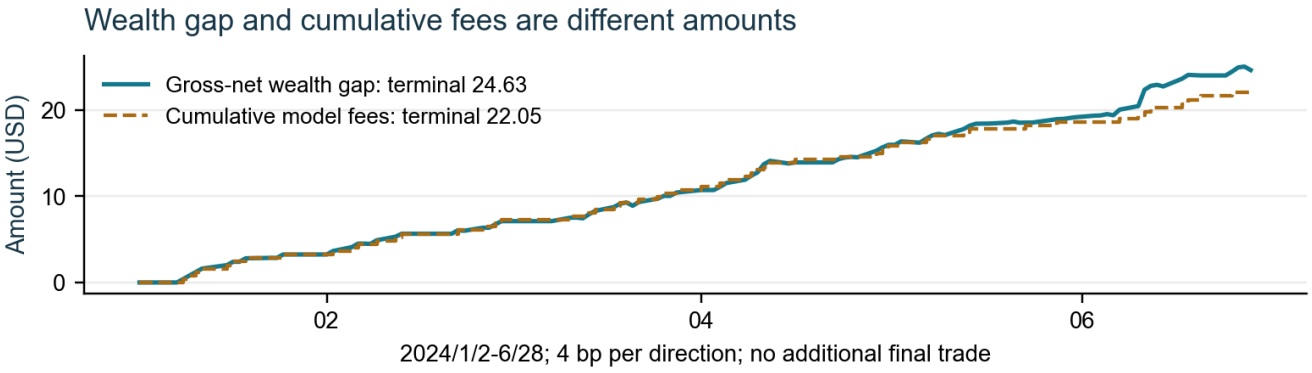
$$W_{end} = W_{start}(1 + r_{net}), \quad 999.60(1 + 0.0052714) \approx 1004.86929$$

Predict: must the terminal gross-net gap equal all dollar fees added together?

```
gross_wealth = 1000*(1+gross).cumprod(skipna=False)
net_wealth = 1000*(1+net).cumprod(skipna=False)
opening = net_wealth.shift(1).fillna(1000)
fees = opening*cost_rate*turnover
pd.DataFrame({"Gross terminal value (USD)": [round(gross_wealth.iloc[-1], 2)],
    "Net terminal value (USD)": [round(net_wealth.iloc[-1], 2)],
    "Model fee sum (USD)": [round(fees.sum(), 2)]})
```

Gross terminal value (USD)	Net terminal value (USD)	Model fee sum (USD)
1134.06	1109.43	22.05

Gross wealth is 1134.06, net 1109.43: a USD 24.63 gap, versus a model fee sum of 22.05. A retained fee would also follow subsequent gross returns. Summing these terminal effects gives 24.63; that later growth explains the extra 2.58. Subsequent declines can instead make the gap smaller than total fees.



The terminal gap is USD 24.63; cumulative model fees are USD 22.05. One terminal deduction cannot replace sequential costs.

Discuss

Gross wealth before buying on 1/9 is USD 1,000. What is the approximate 6/28 value of that USD 0.40 cost following subsequent gross account returns?

Sources: [2], [5].

ISSUE 11

How can USD 1,000 cover both stock and purchase cost?

Calculate an exact purchase budget to locate the approximation, without replacing the main full-period account.

On 1/9 cash is USD 1,000 and C is about USD 185.14/share. q is shares and c=0.0004. The budget must cover both qC and its 4 bp purchase cost.

$$qC + cqC = 1000 \quad \Longrightarrow \quad q = \frac{1000}{C(1+c)}$$

$$q \approx \frac{1000}{185.14(1.0004)} \approx 5.399158 \text{ shares}$$

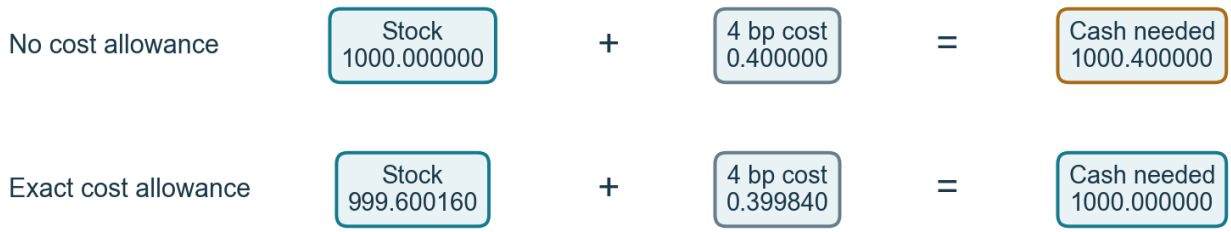
Predict: which row needs more than USD 1,000?

```
q_without = 1000/price
q_reserved = 1000/(price*(1+cost_rate))
counts = np.array([q_without, q_reserved])
pd.DataFrame({"Purchase budget": ["No cost allowance", "Cost reserved"],
    "Shares": counts.round(6),
    "Cash required (USD)": (counts*price*(1+cost_rate)).round(4)})
```

Purchase budget	Shares	Cash required (USD)
No cost allowance	5.401318	1000.4
Cost reserved	5.399158	1000.0

Without an allowance, cash required is 1000.40. Exactly reserving cost gives stock 1000/1.0004≈999.600160 and cost ≈0.399840, totaling 1000. The main approximation records stock 999.600000 and cost 0.400000, each differing by about USD 0.000160: 1/(1+c) is not exactly 1-c. This isolates the first purchase; other issues retain the main approximation.

Exact purchase budget: stock value plus cost equals required cash



All amounts are USD. Exact stock value is 1,000÷1.0004; stock plus cost equals 1,000.

Discuss

The 1/9 main approximation charges USD 0.40; the exact budget charges USD 0.399840. Which amount is each 4 bp charge based on?

Source: [5].

ISSUE 12

How does buying one close later change the first holding return?

Keep the 1/8 signal and change only purchase delay. First compare the actual holding intervals' price changes.

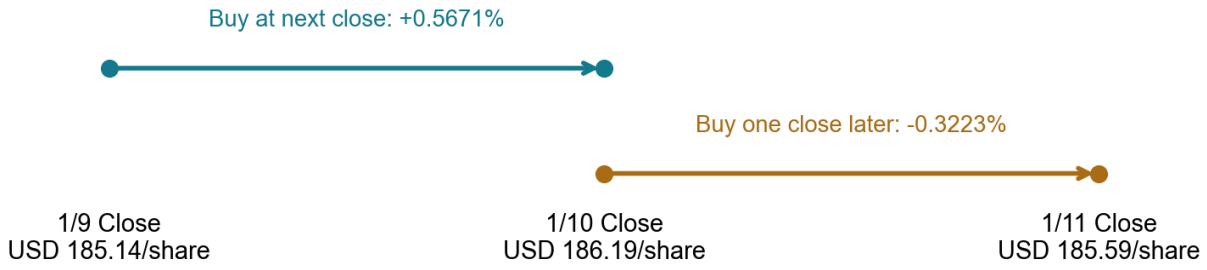
Execution plan	Buy date / price	First end / price
Next close	1/9：185.14	1/10：186.19
One close later	1/10：186.19	1/11：185.59

Prices are USD/share. Fix capital and signal; change only execution delay. The original first return is about +0.5671%; the delayed first return is about -0.3223%. This compares stock prices only, without costs or later trades.

$$\frac{186.19}{185.14} - 1 \approx 0.5671\%, \qquad \frac{185.59}{186.19} - 1 \approx -0.3223\%$$

Delaying execution changes both holding intervals and cost dates. It is not merely moving the result column, or measuring actual slippage from daily data.

Same 1/8 decision: delay changes the first holding dates



The original 1/9 to 1/10 return is positive; the delayed 1/10 to 1/11 return is negative. Later trades and costs are not yet included.

Discuss

If buying only at the 1/10 Close, can the account earn the 1/9 to 1/10 return of +0.5671%?

Sources: [4], [6].

ISSUE 13

Can the first worse return determine the entire delayed account?

Shift every holding interval and cost date together to compare full accounts over the same 1/2 to 6/28 period.

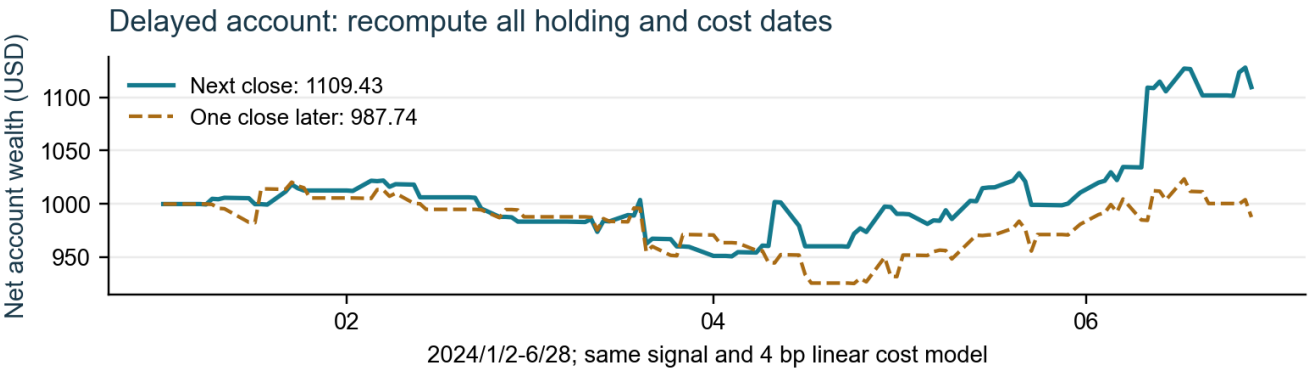
The original account follows the 1/8 signal to buy at the 1/9 Close. The delayed account buys on 1/10; its first return ends on 1/11. Its post-trade state therefore looks back two rows and its holding state three. Both charge 4 bp per direction; the final date is valuation only.

Predict: is a worse first interval enough to calculate the complete terminal gap?

```
delayed_held = signal.shift(3).fillna(0)
delayed_after = signal.shift(2).fillna(0)
delayed_after.iloc[-1] = delayed_held.iloc[-1]
delayed_turnover = (delayed_after-delayed_held).abs()
delayed_gross = asset_return.where(delayed_held.eq(1), 0.0)
delayed_net = delayed_gross-cost_rate*delayed_turnover
delayed_wealth = 1000*(1+delayed_net).cumprod(skipna=False)
pd.DataFrame({"Execution plan": ["Next close", "One close later"],
    "First purchase": ["2024-01-09", "2024-01-10"],
    "Ending net wealth (USD)": [round(net_wealth.iloc[-1], 2),
                        round(delayed_wealth.iloc[-1], 2)]})
```

Execution plan	First purchase	Ending net wealth (USD)
Next close	2024-01-09	1109.43
One close later	2024-01-10	987.74

Terminal wealth is USD 1109.43 originally and 987.74 with delay. Delay also changes every later entry and exit interval. The first interval cannot determine all subsequent results, and this example does not show that every delay reduces return.



Delay changes every later entry and exit, not just the first. These historical paths do not show that all delays produce worse results.

Discuss

Why must cost dates change along with holding dates in the delayed account?

Sources: [4], [6].

ISSUE 14

Do fewer trades necessarily leave more money?

Adopting a new decision every five observations changes both trading frequency and holding intervals. Separate these effects.

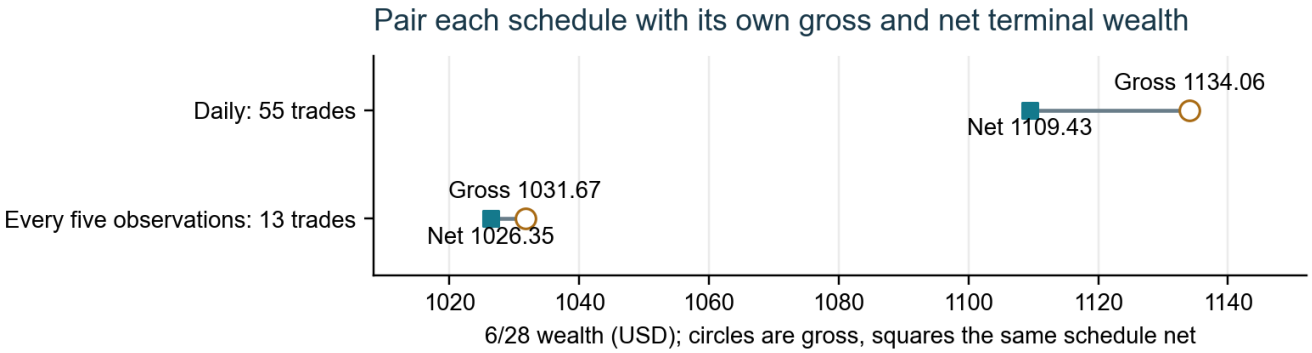
Still calculate the one-day signal daily. Adopt a new decision every five observations from 1/2: 1/2, 1/9, 1/17, and so on; retain it between dates. This is not a five-day return signal or a calendar-week trading schedule.

Predict: with fewer trades, must gross terminal wealth stay the same?

```
scheduled_signal = signal.iloc[::5].reindex(signal.index).ffill()
scheduled_held = scheduled_signal.shift(2).fillna(0)
scheduled_after = scheduled_signal.shift(1).fillna(0)
scheduled_after.iloc[-1] = scheduled_held.iloc[-1]
scheduled_turnover = (scheduled_after-scheduled_held).abs()
scheduled_gross = asset_return.where(scheduled_held.eq(1), 0.0)
scheduled_net = scheduled_gross-cost_rate*scheduled_turnover
pd.DataFrame({"Decision schedule": ["Daily", "Every five observations"],
    "Trade count": [int(turnover.sum()), int(scheduled_turnover.sum())],
    "Gross terminal value (USD)": [round(gross_wealth.iloc[-1], 2),
                      round(1000*(1+scheduled_gross).prod(), 2)],
    "Net terminal value (USD)": [round(net_wealth.iloc[-1], 2),
                      round(1000*(1+scheduled_net).prod(), 2)]})
```

Decision schedule	Trade count	Gross terminal value (USD)	Net terminal value (USD)
Daily	55	1134.06	1109.43
Every five observations	13	1031.67	1026.35

Daily decisions produce 55 trades; every five observations produces 13. The latter ends at gross 1031.67 and net 1026.35, both below daily results. Changed gross wealth shows changed holding periods; the entire difference cannot be attributed to costs.



Fewer trades can still leave less net wealth. Compare gross versus net within a row; a difference across schedules is not solely a cost effect.

Discuss

Which two table values isolate costs without changing holding periods?

Sources: [5], [6].

ISSUE 15

What happens to terminal wealth when only the cost rate rises?

Fix prices, signal, execution, holdings and endpoint before attributing terminal changes to the cost rate.

With positive growth factors and nonnegative costs, increasing the rate cannot increase an interval's factor or their compounded terminal value.

W is wealth in USD; T is 6/28. b is the rate in bp, r_{gross} is gross account return, τ is one-way turnover, and t is each interval's ending date. At the first 1/9 purchase with 10 bp, 1000×(1-0.001)=USD 999.

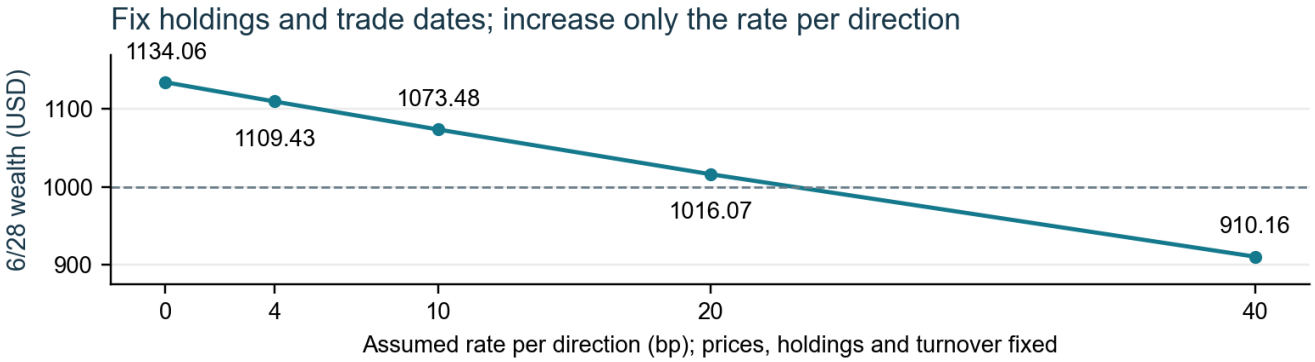
$$W_T(b) = 1000\prod_t (1 + r_{\text{gross},\,t} - \tau_t b/10000)$$

Predict: with other conditions fixed, can terminal wealth at 40 bp exceed that at 4 bp?

```
sensitivity = []
for bp in [0, 4, 10, 20, 40]:
    factors = 1+gross-turnover*bp/10000
    assert factors.gt(0).all()
    sensitivity.append([bp, 1000*factors.prod()])
pd.DataFrame(sensitivity,
    columns=["Rate per direction (bp)", "Terminal wealth (USD)"]).round(2)
```

Rate per direction (bp)	Terminal wealth (USD)
0	1134.06
4	1109.43
10	1073.48
20	1016.07
40	910.16

At 0 bp wealth is 1134.06; at 4 bp, 1109.43; at 40 bp, 910.16. This is sensitivity to specified rates, not actual broker fees identified from daily prices.



A higher rate lowers terminal wealth on this fixed path. The curve does not estimate actual rates from market data.

Discuss

If terminal wealth at 40 bp is higher instead, which holding condition should be checked first?

Sources: [2], [5].

ISSUE 16

Which cost rate eliminates this historical profit?

Find the assumed rate that brings terminal wealth back to USD 1,000, not gross profit divided by trade count.

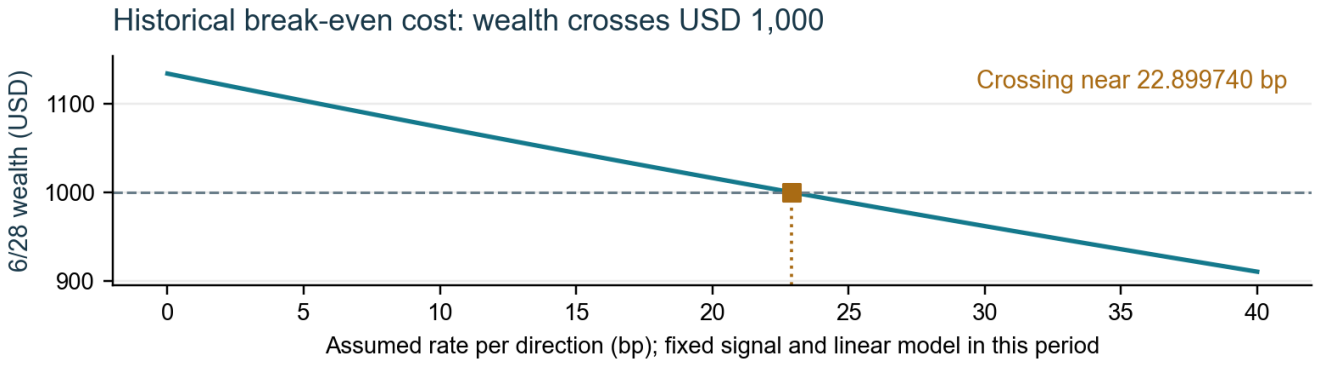
At 0 bp, 1134.06 exceeds 1000; at 40 bp, 910.16 is below 1000. Wealth decreases continuously with rate on this fixed path, so there is one crossing between them. At 20 bp, 1016.07 still exceeds 1000. Keep 20 to 40 and next try 30 bp.

Predict: how do endpoints above and below 1000 help locate the crossing?

```
lo, hi = 0.0, 40.0
assert 1000*(1+gross-turnover*lo/10000).prod() > 1000
assert 1000*(1+gross-turnover*hi/10000).prod() < 1000
for _ in range(50):
    mid = (lo+hi)/2
    value = 1000*(1+gross-turnover*mid/10000).prod()
    if value > 1000:
        lo = mid
    else:
        hi = mid
break_even = (lo+hi)/2
pd.DataFrame({"Break-even rate (bp)": [round(break_even, 6)],
    "Terminal wealth (USD)": [round(value, 5)]})
```

Break-even rate (bp)	Terminal wealth (USD)
22.89974	1000.0

Keep the half containing the crossing each time: about 22.899740 bp per direction. This offsets only the historical profit of this AAPL period, one-day rule, next-close execution and linear model. It is not a future acceptable actual rate.



The crossing belongs only to this AAPL period and account assumptions, not a future acceptable actual rate.

Discuss

If both endpoints exceed 1,000, can a crossing inside the interval be declared?

Sources: [2], [5].

ISSUE 17

How do cash, buy-and-hold and the trading rule compare?

Fix initial USD 1,000 and the 1/2 to 6/28 period. Each holding approach pays its own required trading costs.

Cash holds no stock, trades none and earns no interest. Buy-and-hold buys at the 1/2 Close of USD 185.64/share and pays 4 bp once; the one-day rule pays 55 times. Both value stock at the 6/28 Close of USD 210.62/share without a final sale.

BH denotes buy-and-hold in the formula; W is dollar wealth and T is 6/28. Initial stock value 999.60 still uses the same linear approximation.

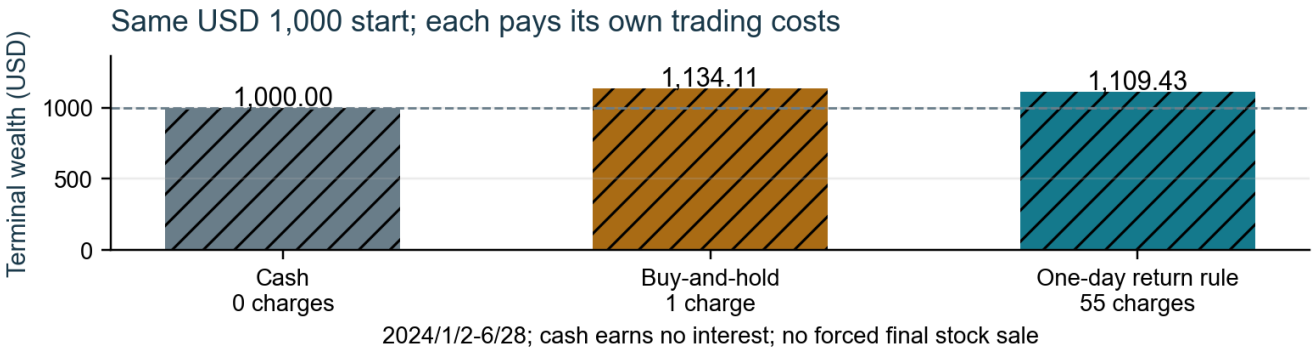
$$W_T^{\mathrm{BH}} = 1000(1-0.0004)\frac{210.62}{185.64} \approx 1134.11 \text{ USD}$$

Predict: is buy-and-hold also charged 4 bp every day?

```
buy_hold_wealth = 1000*(1-cost_rate)*close/close.iloc[0]
pd.DataFrame({"Method": ["Cash", "Buy-and-hold", "One-day return rule"],
    "Charge count": [0, 1, int(turnover.sum())],
    "Terminal wealth (USD)": [1000.0,
        round(buy_hold_wealth.iloc[-1], 2), round(net_wealth.iloc[-1], 2)]})
```

Method	Charge count	Terminal wealth (USD)
Cash	0	1000.0
Buy-and-hold	1	1134.11
One-day return rule	55	1109.43

Buy-and-hold ends at 1134.11 and the main rule at 1109.43. More trades do not imply more money in this period. The 1/2 buy-and-hold cost belongs in wealth, but is an initial-boundary cost, not one of the following 123 daily market returns. Week 7 separates them.



Daily valuation is not daily trading. Buy-and-hold pays only the initial purchase cost; the one-day rule pays for 55 actions.

Discuss

With no final buy-and-hold sale, how many times is this account charged?

Sources: [2], [5].

ISSUE 18

With a USD 1 minimum per order, does a small order still cost only 4bp?

A USD 1 minimum per order can make cost exceed 4bp of traded value.

Separately assume commission of 4bp of traded value with a USD 1 minimum per order. Exclude spread, slippage and tax. This is neither a broker quote nor the main account's composite 4bp cost. Use the 1/9 AAPL Close, USD 185.14/share, without changing the main account.

q is order shares, C is USD/share, and F is commission in USD. max selects the larger amount. Divide F by stock traded value qC, then multiply by 10,000 to express the rate in bp.

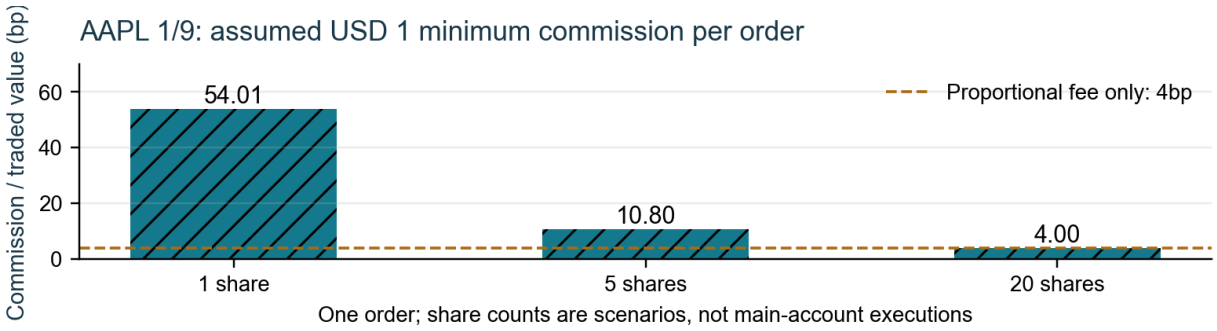
$$F = \max(1, 0.0004qC), \qquad \text{fee in bp} = \frac{F}{qC}10000$$

For one share, 0.0004×185.14≈USD 0.07406, replaced by the minimum 1. Then 1÷185.14×10,000≈54.01bp. Share counts only test the fee schedule; 20 shares are not a main-account order.

Predict: under this same fee schedule, do one and five shares have the same cost rate in bp?

```
order_units = np.array([1, 5, 20])
order_amount = order_units*price
order_fee = np.maximum(1.0, .0004*order_amount)
order_bp = order_fee/order_amount*10000
pd.DataFrame({"Shares": order_units, "Traded value (USD)": order_amount.round(2),
    "Commission (USD)": order_fee.round(5), "Cost rate (bp)": order_bp.round(2)})
```

Shares	Traded value (USD)	Commission (USD)	Cost rate (bp)
1	185.14	1.0	54.01
5	925.7	1.0	10.8
20	3702.8	1.48112	4.0



Five shares still cost USD 1, or 10.80bp; 20 shares pay 4bp. The stock-value threshold is 1÷0.0004=USD 2,500, requiring USD 2,501 cash including cost. Splitting small orders repeats the minimum charge.

Discuss

With a separate USD 1,000 cash budget, what is the maximum ideal fractional purchase in one order under this fee schedule? Why is Issue 11's proportional-only budget inapplicable?

Sources: [2], [5].

References

Price sources and reproduction conditions

[1] yfinance: historical price fields

[2] Investor.gov: understanding fees

[3] Investor.gov: ask price and bid-ask spread

[4] Investor.gov: executing an order

[5] Lobo, Fazel and Boyd: proportional and fixed transaction costs

[6] pandas: shifting observations

AAPL daily data: 2024/1/2 to 6/28, 124 observations from the Yahoo/yfinance snapshot retained on 2026/9/14. Trading rules, execution and costs are teaching assumptions, not actual execution records.

Prices display 2 decimal places; calculations retain source precision. Reproduce with the recorded data, without replacing results by a new download.

Python 3.12.14；numpy 2.2.6；pandas 2.2.3；matplotlib 3.10.8；yfinance 1.2.0。

Week 7 continues the original composite 4bp linear-approximation account, USD 1109.43, not Issue 18's minimum commission schedule.